

Algebra PhD Exam, May 1996

Answer seven out of the following ten problems. State clearly any results you need.

1. Let R be an integral domain, with fraction field K . Recall that an R -submodule I of K is said to be invertible if $II^{-1} = R$, where $I^{-1} = \{x \in K \mid ix \in R \text{ for all } i \in I\}$. Show that if I is invertible, then it is finitely generated and projective.
2. This problem has two parts.
 - (a) Let k be an algebraically closed field and R a commutative, finitely generated k -algebra. Show that the Jacobson radical of R is the same as the nilradical. (Hint: Nullstellensatz).
 - (b) Give an example of a commutative ring with identity whose Jacobson radical is not equal to its nilradical.
3. Compute the Galois group of the extension $K/\mathbb{Q}$, where K is the splitting field over $\mathbb{Q}$ of the polynomial $X^4 - 3$.
4. Let K be a field, and R a simple, finite-dimensional K -algebra with identity (not necessarily commutative). Show that if the identity is the only (nonzero) idempotent element of R , then R is a division ring.
5. Let R be a discrete valuation ring with fraction field K . Show that R is integrally closed in K .
6. This problem has two parts.
 - (a) Suppose G is a finite p -group and S is a finite set on which G acts. If S^G is the set of elements of S fixed by G , show that $|S| \equiv |S^G| \pmod{p}$.
 - (b) Suppose G is a p -group, k is a field of characteristic p , V is a finite-dimensional k -vector space, and $\rho : G \rightarrow \text{GL}(V)$ is a homomorphism. Show that there is a nonzero vector $v \in V$ such that $\rho(g)(v) = v$ for all $g \in G$. Hint: reduce to the case where k is finite, and apply (a).
7. Let k be a field and $G = \text{GL}(3, k)$. Describe the conjugacy classes of elements of order 3 in G if
 - (a) $k = \mathbb{C}$,
 - (b) $k = \mathbb{R}$,
 - (c) $k = \mathbb{F}_3$.
8. Let R be a commutative ring. Define what it means for an R -module to be injective. Show that an R -module M is injective if and only if for every ideal $I \subseteq R$ and R -linear map $f : I \rightarrow M$, f extends to a map $R \rightarrow M$.
9. This problem has two parts.
 - (a) Show that there exist solvable groups of arbitrarily large derived length.
 - (b) Show that if S is an infinite set, then there is no “free solvable group generated by S ”; i.e. there is no solvable group F and map $\alpha : S \rightarrow F$ such that composition with α induces, for any solvable group G , a bijection between the set of group homomorphisms $F \rightarrow G$ and the set of maps $S \rightarrow G$.
10. State and prove the Hilbert Basis Theorem.